



KOLESA.KZ

PREMIUM

AUTO MARKET ANALYSIS

K.A.T.A TEAM

WE HAVE COLLECTED

12000+

Raw Listings

60M+

KZT Treshhold

17

Cities covered

INTRODUCTION

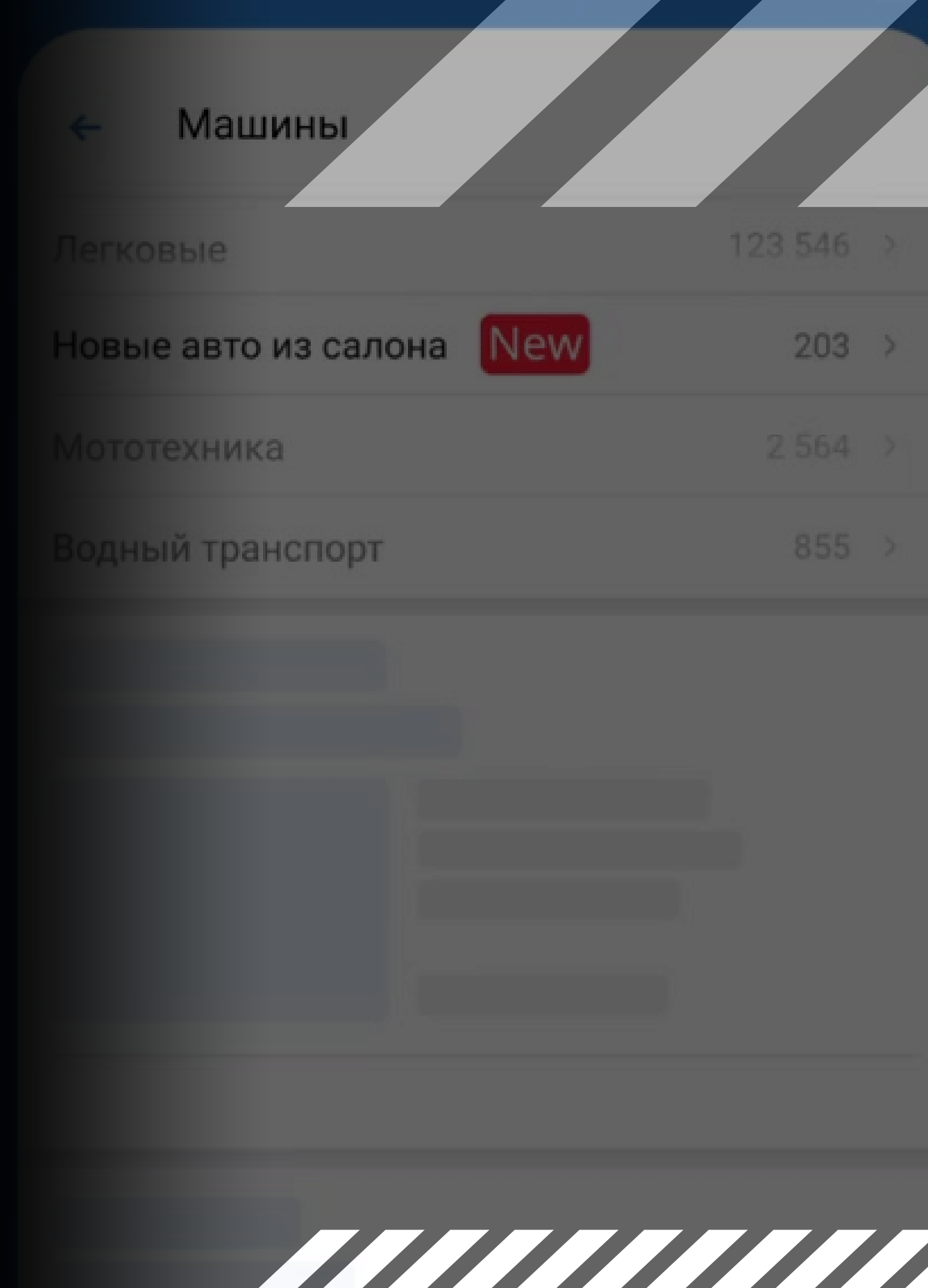
WEBSITE

KOLESA.KZ

Kazakhstan's largest automotive marketplace

TARGET SEGMENT

Cars listed above 60,000,000 KZT — the Premium & Luxury tier of Kazakhstan's auto market.



PROJECT GOAL

■ Price Anomaly Detection

Compare *actual vs. mean* market price to identify undervalued or overpriced luxury assets

■ Regional Trends

Analyze concentration of premium listings across cities: **Almaty, Astana**, and beyond

■ Depreciation Analysis

Measure how mileage and age impact vehicle value in the **60M+ KZT** bracket


```
import time
import requests
import random
import pandas as pd
from selenium import webdriver
from selenium.webdriver.chrome.options import Options
from selenium.webdriver.chrome.service import Service
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
from bs4 import BeautifulSoup
from webdriver_manager.chrome import ChromeDriverManager
```

DATA COLLECTION

```
BASE_URL = "https://kolesa.kz"
PRICE_FROM = 60_000_000
LIST_WAIT = 15
DETAIL_WAIT = 15
PAGE_SLEEP = (1.5, 2.5)
CARD_SLEEP = (2.0, 3.5)
SCROLL_SLEEP = 0.8
```

```
CITIES = {
    "all": "",
    # "almaty": "almaty",
    # "astana": "astana",
    # "shymkent": "shymkent",
    # "karaganda": "karaganda",
    # "aktobe": "aktobe",
    # "taraz": "taraz",
    # "pavlodar": "pavlodar",
    # "oskemen": "oskemen",
    # "semev": "semev",
}
```

TOOLS USED

SELENIUM

Browser automation & dynamic
content loading

BEAUTIFULSOUP

HTML parsing & data extraction

PANDAS

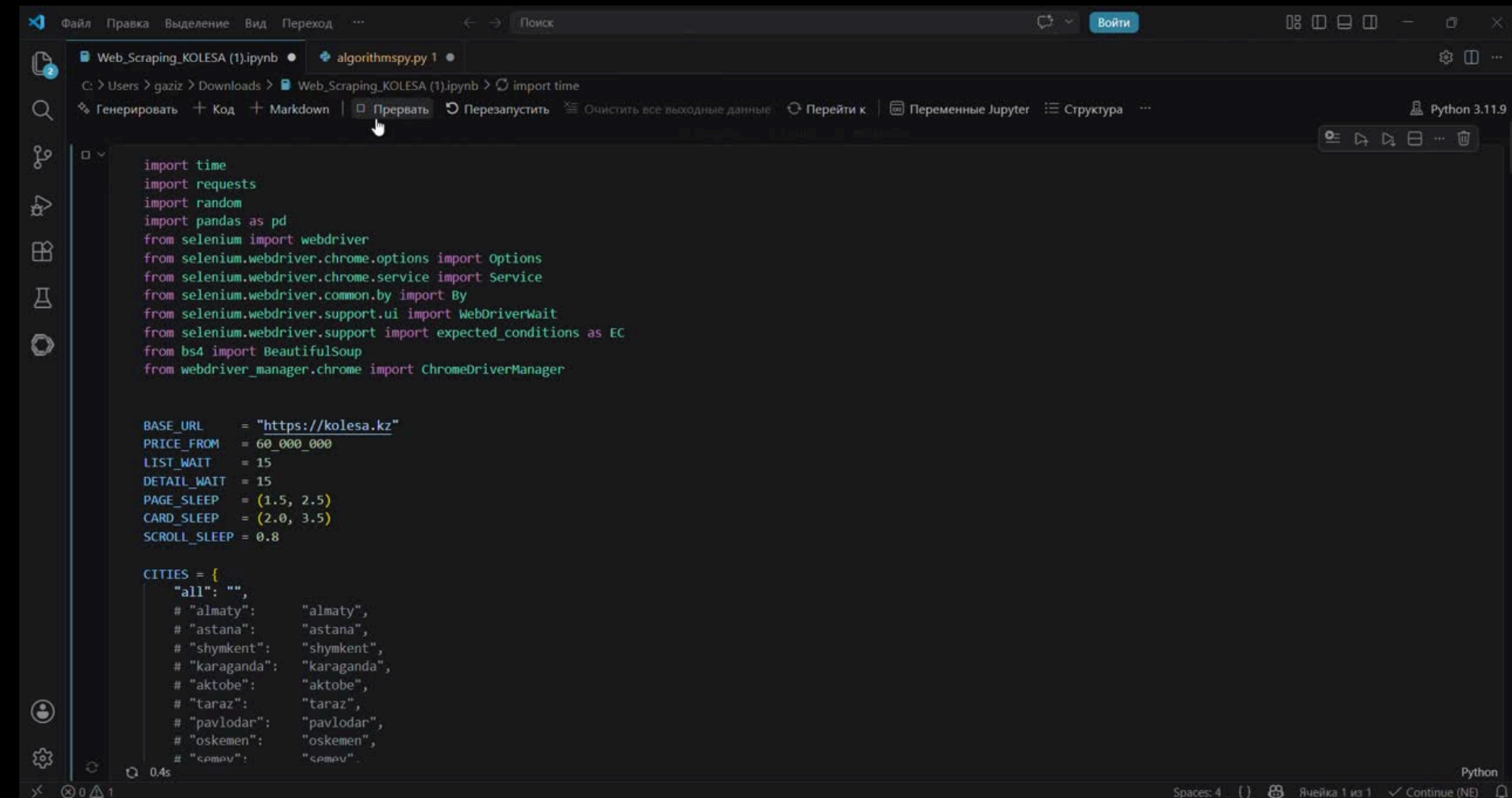
Data aggregation & CSV export

REQUESTS

Site availability checks & HTTP

SCRAPING PROCESS

- 1 Set price filter: $\geq 60,000,000$ KZT
- 2 Scroll & paginate through listing pages
- 3 Extract card data (title, price, city, year)
- 4 Visit each listing for detail fields
- 5 Parse mean_price from window.digitalData
- 6 Export: 1,200 rows → Kolesa_Raw_Data.csv



```
import time
import requests
import random
import pandas as pd
from selenium import webdriver
from selenium.webdriver.chrome.options import Options
from selenium.webdriver.chrome.service import Service
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
from bs4 import BeautifulSoup
from webdriver_manager.chrome import ChromeDriverManager

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    # "astana": "astana",
    # "shymkent": "shymkent",
    # "karaganda": "karaganda",
    # "aktobe": "aktobe",
    # "taraz": "taraz",
    # "pavlodar": "pavlodar",
    # "oskemen": "oskemen",
    # "semey": "semey"
```

Challenges: Anti-bot detection → randomized delays (1.5–3.5s) · Dynamic JS content → Selenium wait conditions · Rate limiting → exponential backoff

STEPS PERFORMED

● Handle Missing Values

- Dropped 3 columns >91% null
- Filled mileage **NaN** → **0** (brand new)
- Filled *mean_price* with median

● Remove Duplicates

- Step A: Exact URL-based deduplication
- Step B: Content deduplication (*title + year + price + city*)

● Fix Data Types

- Price: string → integer (strip ₹ symbol)
- Mileage: string → integer
- Engine volume: extract float

● Clean Text

- Strip whitespace & hidden chars
- Normalize to Title Case for consistent *GROUP BY* in SQL

● Feature Engineering

- *brand*: first word of title
- *price_diff*: $\text{mean_price} - \text{price}$
- *is_bargain*: $\text{price} < 95\%$ of mean

● New Computed Fields

- *car_age* = $2026 - \text{year}$
- *price_per_year* = $\text{price} / (\text{age} + 1)$
- Output → **Kolesa_Final_Cleaned.csv**

DATA PREPROCESSING

BEFORE

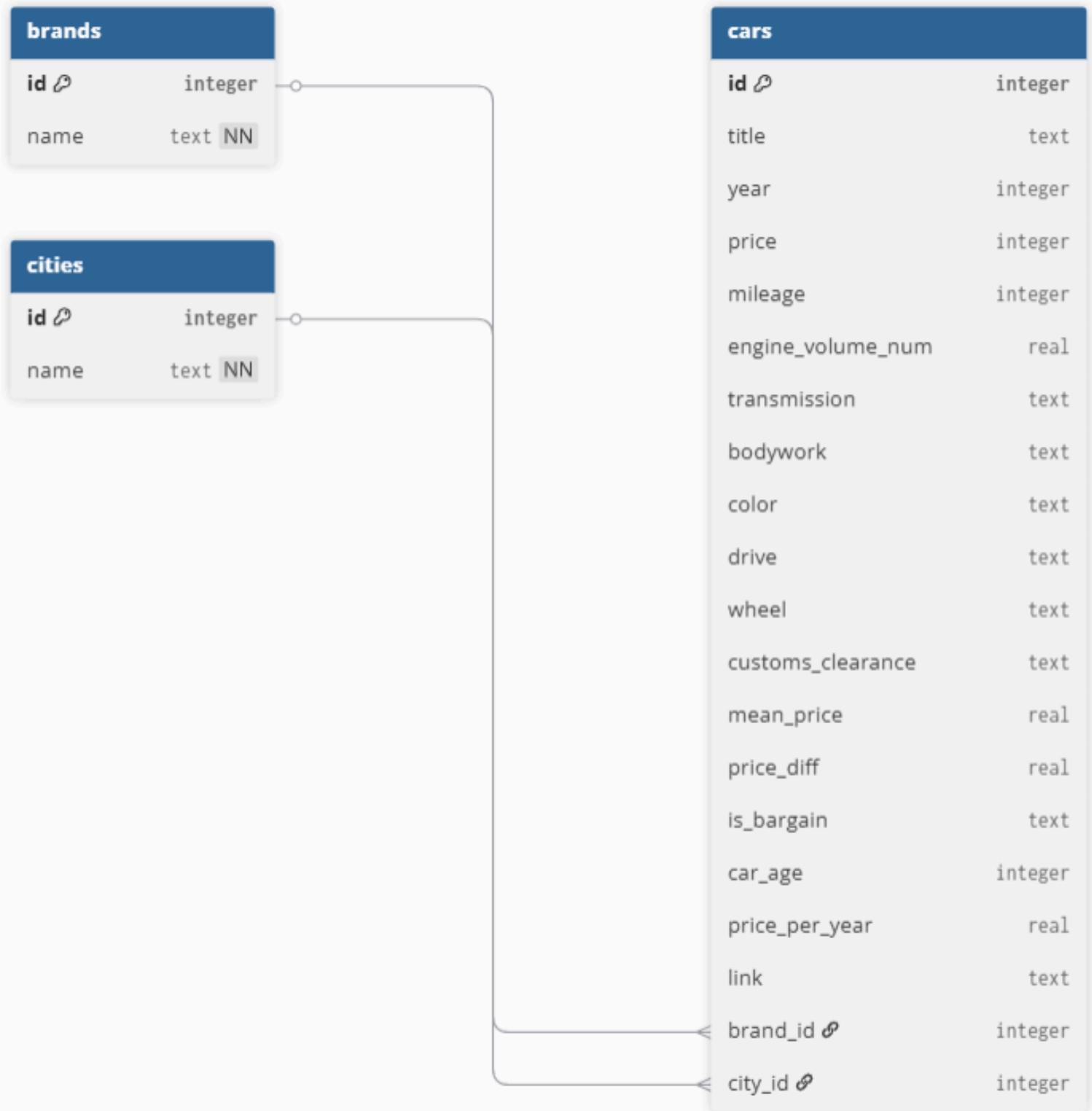
1200 rows

	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1		price	city	date	views	is_verified_seller	is_from_dealer	is_official_dealer	mean_price	offer_price	generation	bodywork	engine_volume	mileage	transmission
2	kolesa.kz/a/bhc	65000000	Кокшетау	30 апреля	223	FALSE	FALSE	FALSE	53224000	65 000 000 ₸	2018 - н.в. 5 поколение	Пикап	3 (бензин)	2000	Автомат
3	kolesa.kz/a/bhc	77500000	Астана	30 апреля	573	TRUE	FALSE	FALSE	86302000	77 500 000 ₸	2021 - н.в. 4 поколение	Внедорожник	3.5 (бензин)	43017	Автомат
4	kolesa.kz/a/bhc	77493000	Павлодар	30 апреля	494	FALSE	FALSE	TRUE	65066000	77 493 000 ₸	2023 - н.в. G05 рест	Кроссовер	3 (бензин)	0	Автомат
5	kolesa.kz/a/bhc	71738000	Павлодар	30 апреля	3065	FALSE	FALSE	TRUE	65066000	71 738 000 ₸	2023 - н.в. G05 рест	Кроссовер	3 (бензин)	0	Автомат
6	kolesa.kz/a/bhc	74000000	Астана	30 апреля	1801	FALSE	FALSE	FALSE	73055000	74 000 000 ₸	2021 - н.в. 4 поколение	Внедорожник	3.5 (бензин)	47000	Автомат
7	kolesa.kz/a/bhc	71738000	Семей	30 апреля	1799	FALSE	FALSE	TRUE	65066000	71 738 000 ₸	2023 - н.в. G05 рест	Кроссовер	3 (бензин)	0	Автомат
8	kolesa.kz/a/bhc	110000000	Астана	30 апреля	827	FALSE	FALSE	FALSE	100000000	110 000 000 ₸	2024 - н.в. 5 поколение	Внедорожник	6.2 (бензин)	10000	Автомат
9	kolesa.kz/a/bhc	69590000	Алматы	30 апреля	323	TRUE	FALSE	FALSE	71561000	69 590 000 ₸	2022 - н.в. G07 рест	Кроссовер	3 (бензин)	21683	Автомат
10	kolesa.kz/a/bhc	81327000	Алматы	30 апреля	567	TRUE	FALSE	FALSE	79103000	81 327 000 ₸	2019 - н.в. 2 поколение	Внедорожник	3 (бензин)	20	Автомат
11	kolesa.kz/a/bhc	76990000	Караганда	30 апреля	1193	FALSE	FALSE	TRUE	79103000	76 990 000 ₸	2024 - н.в. Z63	Внедорожник	3.5 (бензин)	0	Автомат
12	kolesa.kz/a/bhc	110000000	Алматы	30 апреля	2142	FALSE	FALSE	FALSE	79103000	110 000 000 ₸	2020 - н.в. W223	Седан	4 (гибрид)	17000	Автомат
13	kolesa.kz/a/bhc	124000000	Астана	30 апреля	7257	FALSE	FALSE	TRUE	79103000	124 000 000 ₸	2019 - н.в. X167	Внедорожник	4 (бензин)	0	Автомат
14	kolesa.kz/a/bhc	96641305	Алматы	30 апреля	1978	FALSE	FALSE	TRUE	79103000	96 641 305 ₸	2022 - н.в. G07 рест	Кроссовер	4.4 (бензин)	6	Автомат
15	kolesa.kz/a/bhc	85000000	Алматы	30 апреля	3045	FALSE	FALSE	FALSE	79103000	85 000 000 ₸	2019 - н.в. 2 поколение	Лифтбек	4 (бензин)	20000	Автомат
16	kolesa.kz/a/bhc	74890000	Астана	30 апреля	3341	FALSE	FALSE	TRUE	67236000	74 890 000 ₸	2021 - н.в. J300	Внедорожник	3.5 (бензин)	0	Автомат
17	kolesa.kz/a/bhc	110000000	Алматы	30 апреля	834	FALSE	FALSE	FALSE	113951000	110 000 000 ₸	2018 - н.в. W463	Внедорожник	4 (бензин)	57500	Автомат
18	kolesa.kz/a/bhc	117000000	Петропавловск	30 апреля	750	FALSE	FALSE	FALSE	79103000	117 000 000 ₸	2019 - н.в. 1 поколение	Минивэн	Не указан	7000	Автомат
19	kolesa.kz/a/bhc	144500000	Алматы	30 апреля	143	FALSE	FALSE	FALSE	79103000	144 500 000 ₸	2020 - н.в. W223	Седан	4 (гибрид)	0	Автомат
20	kolesa.kz/a/bhc	67000000	Алматы	30 апреля	927	FALSE	FALSE	FALSE	69580000	67 000 000 ₸	2021 - н.в. 4 поколение	Внедорожник	3.5 (бензин)	85000	Автомат
21	kolesa.kz/a/bhc	90000000	Алматы	30 апреля	2085	FALSE	FALSE	FALSE	79103000	90 000 000 ₸	2014 - 2020 1 покол	Седан	6.6 (бензин)	13000	Автомат
22	kolesa.kz/a/bhc	94290000	Павлодар	30 апреля	692	FALSE	FALSE	TRUE	79300000	94 290 000 ₸	2022 - н.в. G07 рест	Кроссовер	3 (бензин)	0	Автомат
23	kolesa.kz/a/bhc	88990000	Алматы	30 апреля	4316	FALSE	FALSE	TRUE	79103000	88 990 000 ₸	2024 - н.в. Z63	Внедорожник	3.5 (бензин)	0	Автомат
24	kolesa.kz/a/bhc	60000000	Алматы	30 апреля	1049	FALSE	FALSE	FALSE	60175000	60 000 000 ₸	2020 - н.в. F90 рест	Седан	4.4 (бензин)	74000	Автомат
25	kolesa.kz/a/bhc	74306120	Алматы	30 апреля	4046	FALSE	FALSE	TRUE	79103000	74 306 120 ₸	2014 - н.в. W447	Минивэн	2 (дизель)	5	Автомат
26	kolesa.kz/a/bhc	159000000	Алматы	30 апреля	10800	TRUE	FALSE	FALSE	79103000	159 000 000 ₸	2019 - н.в. 1 поколение	Внедорожник	4 (гибрид)	0	Автомат
27	kolesa.kz/a/bhc	101220000	Алматы	30 апреля	856	FALSE	FALSE	TRUE	79103000	101 220 000 ₸	2024 - н.в. 2 поколение	Кроссовер	Не указан	0	Автомат
28	kolesa.kz/a/bhc	71000000	Астана	30 апреля	841	FALSE	FALSE	FALSE	69375000	71 000 000 ₸	2020 - н.в. W223	Седан	3 (бензин)	23000	Автомат
29	kolesa.kz/a/bhc	83300000	Костанай	30 апреля	2684	FALSE	FALSE	FALSE	79103000	83 300 000 ₸	2021 - н.в. 4 поколение	Внедорожник	3.5 (бензин)	19000	Автомат
30	kolesa.kz/a/bhc	68300000	Алматы	30 апреля	2548	FALSE	FALSE	FALSE	67236000	68 300 000 ₸	2021 - н.в. J300	Внедорожник	3.5 (бензин)	10500	Автомат

AFTER

750~ rows
3 cols removed
4 cols added

DATABASE DESIGN



DESIGN
DECISIONS

NORMALIZATION

brands & cities extracted to separate tables — prevents redundancy and enables fast GROUP BY JOINS

FOREIGN KEYS

brand_id and city_id link cars to dimension tables — ensures referential integrity

DERIVED COLS

price_diff, is_bargain, car_age computed at load time — accelerates analytical queries

SQLITE ENGINE

Lightweight, zero-configuration — ideal for academic analysis via Python's sqlite3 module

SQL QUERIES & INSIGHTS

1 Average Price by Brand

```
query1 = ""
SELECT b.name as Brand, ROUND(AVG(c.price), 0) as Avg_Price, COUNT(c.id) as Total_Cars
FROM cars c
JOIN brands b ON c.brand_id = b.id
GROUP BY b.name
HAVING Total_Cars > 5
ORDER BY Avg_Price DESC
LIMIT 10
""
```

Ferrari & Rolls-Royce lead at 160M+ KZT average — 3× more than Mercedes-Benz entry models

2 Listings Distribution by City

```
query2 = ""
SELECT ci.name as City, COUNT(c.id) as Listing_Count
FROM cars c
JOIN cities ci ON c.city_id = ci.id
GROUP BY ci.name
ORDER BY Listing_Count DESC
""
```

Almaty (51%) + Astana (24%) = 75% of all premium inventory in Kazakhstan

3 Top High-End Listings

```
query3 = ""
SELECT title, price, year, mileage
FROM cars
WHERE price > 50000000 AND mileage < 20000
ORDER BY price DESC
LIMIT 10
""
```

Identifies 'unicorns' — near-new super-luxury cars; 2024 models dominate the top 10

4 Bargain Hunting by City

```
query4 = ""
SELECT ci.name as City, COUNT(c.id) as Bargain_Count, ROUND(AVG(c.price_diff), 0) as Avg_Savings
FROM cars c
JOIN cities ci ON c.city_id = ci.id
WHERE c.is_bargain = 'Good Price' OR c.is_bargain = 'HOT DEAL'
GROUP BY ci.name
ORDER BY Avg_Savings DESC
""
```

Certain regional markets (Karaganda, Shymkent) show higher below-average pricing

5 Inventory Age Distribution

```
query5 = ""
SELECT
  CASE
    WHEN year >= 2020 THEN 'New (2020-2026)'
    WHEN year >= 2010 THEN 'Modern (2010-2019)'
    WHEN year >= 2000 THEN 'Older (2000-2009)'
    ELSE 'Vintage (<2000)'
  END as Era,
  COUNT(*) as Car_Count,
  AVG(price) as Avg_Era_Price
FROM cars
GROUP BY Era
ORDER BY year DESC
""
```

76% of listings are 2020+ models; rare classics (pre2010) maintain high collector value

VISUALIZATION

GOOGLE SHEETS

Interactive pivot tables and charts built directly from Kolesa_Cleaned_Version_Sheets.xlsx for quick ad-hoc analysis.

- Dynamic pivot tables
- Conditional formatting for bargains
- City breakdown charts

Google Docs

Google Sheets

Google Slides

Google Forms

Google Maps

Google Drive

Google Calendar

Google Classroom

Google Meet

Google Assistant

Google Translate

Google Lens

Google Assistant

Google Assistant

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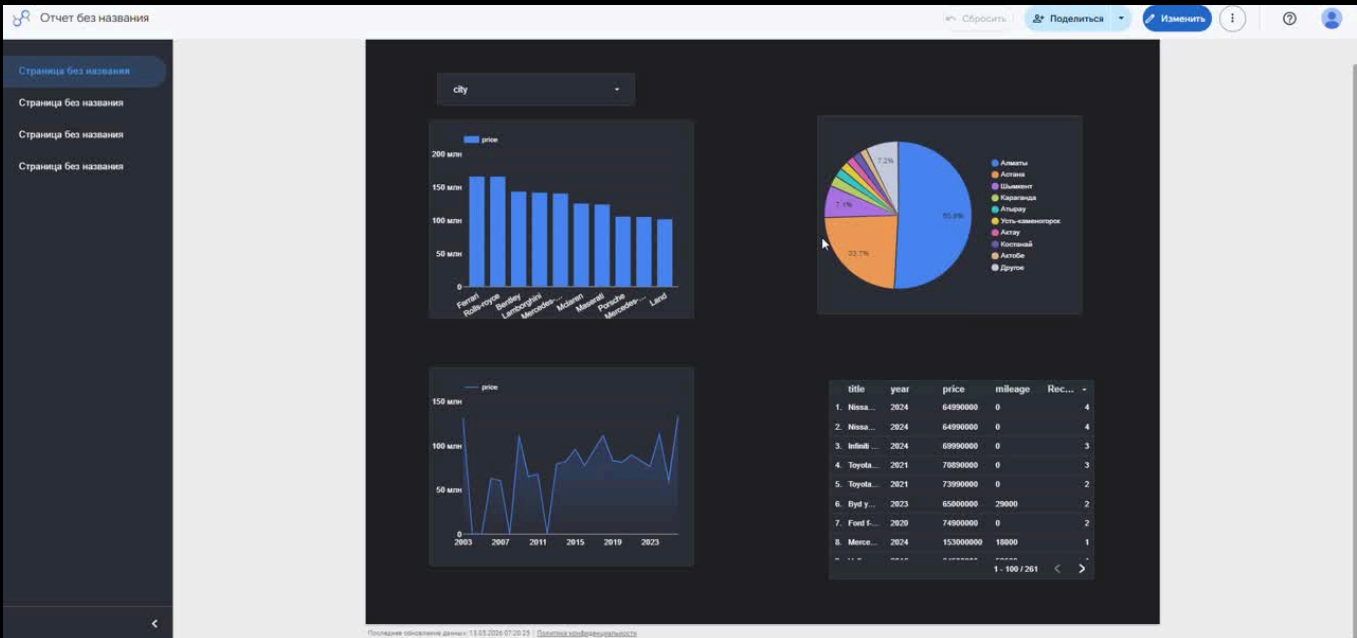
Google Assistant

Google

LOOKER STUDIO

Full business intelligence dashboard with real-time filtering by city, brand, and price range.

- City pie chart (Almaty 50.8%)
- Brand avg-price bar chart
- Year-price trend line
- Sortable data table



KEY DASHBOARD INSIGHTS

GEOGRAPHY

50.8% Almaty · 23.7% Astana → 74.5%
market concentration in 2 cities

BRAND PEAK

Ferrari & Rolls-Royce avg > 160M KZT — 2–
3× higher than Porsche & Mercedes-Benz

VALUE TRAP

Pre-2010 models spike above 100M KZT —
rare classics defy typical depreciation
curve

TOP-3 MOST EXPENSIVE CARS



Porsche 911 GT3 RS
310 000 000



Rolls-Royce Cullinan
470 000 000



Mercedes-Benz G 63 AMG
285 000 000

TOP-3 MOST RAREST CARS IN KAZAKHSTAN



Maserati MC20

160 000 000



Ferrari Purosangue

225 000 000



McLaren 570S

125 000 000

TOP-3 MOST EXPENSIVE OLDEST CARS



Ferrari 458 2009

145 000 000



Rolls-Royce Phantom 2003

200 000 000



Rolls-Royce Wraith 2012

100 000 000

TOP-3 MOST EXPENSIVE NEWEST CARS



Mercedes-Benz G 63 AMG 2025

238 000 000



Mercedes-Maybach S 680 2026

170 000 000



Porsche 911 GT3 2025

216 127 800

TOP 3 CARS WITH THE LONGEST MILEAGE



Mercedes-Benz S 280 1995

100 000 000

270 000 km



Toyota Camry 2013

65 000 000

313 755 km



Audi RS7 2019

73 500 000

173 850 km

TOP-3 MOST POPULAR CARS



Mercedes-benz gls 63 amg

67 500 000

37 889 views



Porsche 911 GT3 RS

310 000 000

39 633 views



Tesla Cybertrack

95 000 000

37 751 views

CONCLUSION

KEY FINDINGS

Market is geographically concentrated —
Almaty & Astana account for ~75% of all
60M+ KZT listings

Ferrari, Rolls-Royce, Bentley &
Lamborghini form an ultra-premium
cluster averaging above 130M KZT

Majority (76%) of premium listings are
2020+ models with 0 km, indicating a
healthy new-car market

Pre-2010 cars spike in price — collectors'
vehicles defy standard depreciation curves

~15% of listings qualify as 'bargains' (priced
≥5% below market mean), mainly in
secondary cities

LIMITATIONS

1. Snapshot data — listing prices fluctuate daily

2. ~37% records dropped as duplicates/outliers

3. mean_price sourced from Kolesa's own algorithm,
not independent valuation

4. No historical time-series — single scrape
prevents trend analysis over time

K.A.T.A TEAM

THANK YOU

• This project demonstrates an end-to-end data pipeline: web scraping → preprocessing → SQL database → Sheets → LS visualization — applied to Kazakhstan's premium automotive sector.

KENZHALY GAZIZ

SHAPAY DAMIR

AKHMET ALIKHAN

